TIME SERIES ANALYSIS AND FORECASTING · UCLA ANDERSON · DR. WILLIAM YU

Seeing the Surge Coming

AI vs. statistical forecasts of U.S. flu activity, deployed as a live dashboard

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October 22, 2026



THE QUESTION

Can we see a flu surge coming one to four weeks ahead?

User

A national or regional public-health planner

Decision

When to scale up staffing, supplies and public messaging

Forecast output

The probability that flu activity crosses a high-activity threshold within 1–4 weeks

Target measure: weekly U.S. ILI+ (influenza-like illness “plus”), the estimated share of outpatient doctor visits that are flu

CHOOSING THE TARGET

Why ILI+: flu-specific and sized to the season

ILI+ = % of outpatient visits for influenza-like illness × % of flu tests that are positive

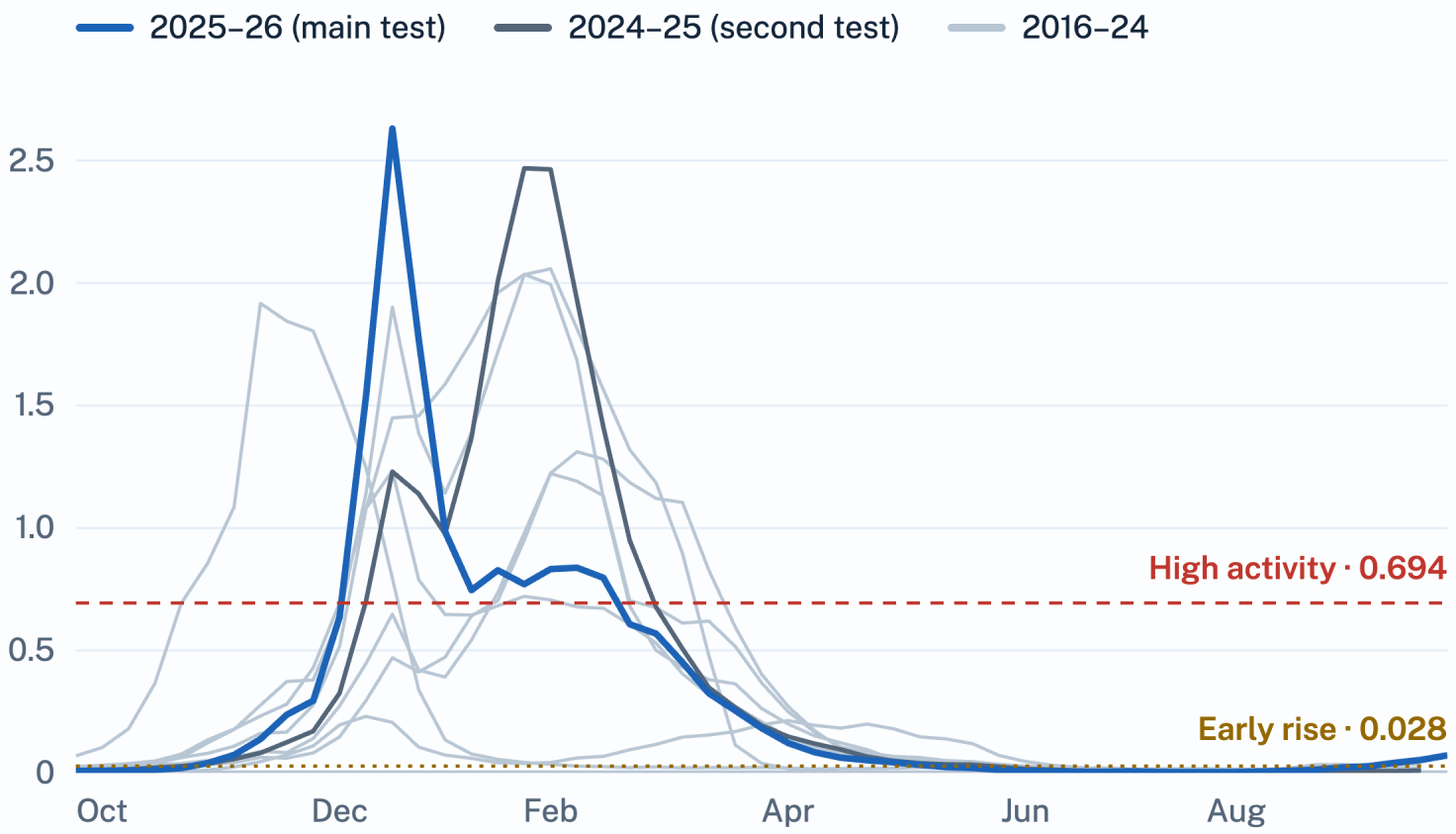
Candidate	What it measures	Why not on its own
Test positivity	Share of tested patients with flu	Depends on who gets tested; no sense of scale
ILI %	Share of visits with flu-like symptoms	Also counts COVID-19, RSV and colds
Hospital admissions	Severe flu burden	Consistent only since ~2021–22: too short
Wastewater	Virus shed in sewage	Influenza A only; about four seasons of data
ILI+ (chosen)	Share of visits that are flu	Flu-specific and scaled; weekly since 2016–17

2021–22: ILI peaked at 4.9% because of COVID-19, but ILI+ peaked at only 0.23. 2023–24 positivity peaked at 18% vs. 25% in 2016–17, yet both ILI+ peaks were 1.2.

Source: CDC FluView (ILINet, clinical labs) via Delphi Epidata API

THE DATA

Ten seasons of ILI+ and two thresholds



523 weeks

Oct 2016 – Oct 2026, national, no gaps

1.2 – 2.6

Peak ILI+ in normal seasons

≈ 0

2020–21: flu nearly absent; imputed for modeling

Thresholds set from pre-test seasons only, before results were seen.

ILI+ by week of flu season (MMWR week 40 = early October). Source: CDC FluView via Delphi Epidata API

METHODS

Five models, one evaluation design

Baseline

Naive

Same week last year

Statistical

ETS

Exponential smoothing: error, trend, seasonality

Statistical

ARIMA

Autoregressive model with yearly sine and cosine terms

Statistical

Prophet

Meta's additive trend and seasonality model

AI

Chronos

Pretrained AI forecaster, used zero-shot

Rolling origin

Refit every week on past data only

Horizons

1–4 weeks ahead; 500 simulated paths

Test seasons

Main: 2025–26
Check: 2024–25

COVID-19

2020–21 imputed; 2021–22 kept as observed

Metrics: MAE (average miss) · CRPS (accuracy of the whole forecast range) · 90% interval coverage · Brier score (accuracy of probabilities)

RESULTS · FORECAST ACCURACY

ARIMA was the most consistent; Chronos was accurate but overconfident

Model	CRPS 2025–26	CRPS 2024–25	MAE 2025–26	MAE 2024–25	90% coverage
Naive (baseline)	0.253	0.275	0.355	0.319	0.96 / 0.85
ETS	0.227	0.181	0.289	0.248	0.81 / 0.86
ARIMA	0.184	0.160	0.234	0.211	0.87 / 0.86
Prophet	0.150	0.285	0.219	0.391	0.97 / 0.86
Chronos (AI)	0.200	0.201	0.225	0.226	0.60 / 0.61 low

ARIMA: 27–42% lower CRPS than naive in both seasons, and best at 1 week ahead

Prophet: best in 2025–26, but worse than naive in 2024–25

Chronos: steady, but its 90% ranges held the actual value only ~60% of the time

CRPS: continuous ranked probability score. MAE: mean absolute error. Lower is better; 1–4 week average. Blue = best per column.

RESULTS · SURGE PROBABILITIES

Only Prophet beat the baseline on surge probability in both seasons

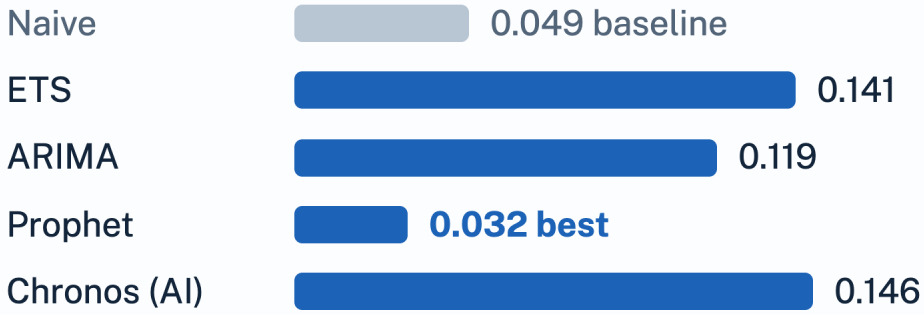
2025–26

4 of 24 scored weeks saw a crossing



2024–25

4 of 22 scored weeks saw a crossing



Event: ILI+ rises above 0.694 within 1–4 weeks (scored while below it)

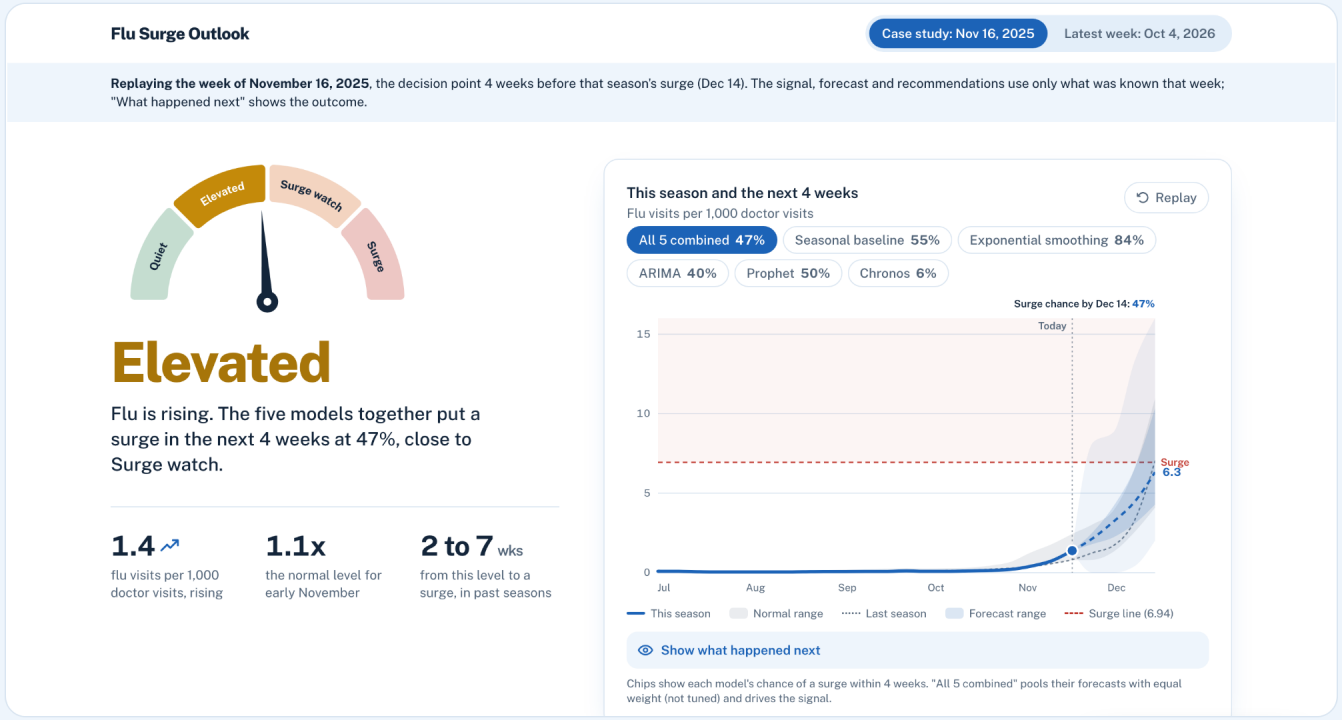
Every model beat naive in 2025–26; **only Prophet** did in 2024–25

With four crossings per season, these rankings are **fragile**

Brier score: mean squared error of predicted probabilities; lower is better. Bar length is proportional to the score.

PRODUCT DEMO

From forecast to product: the Flu Surge Outlook



- Status at a glance: Quiet, Elevated, Surge watch, Surge
- Surge chance from each of the five models, plus a combined forecast
- Actions for hospitals, public health, clinics and families
- State map of where flu is rising first
- AI assistant answers questions from the page's numbers

The dashboard shows flu visits per 1,000 doctor visits (ILI+ × 10), so its surge line of 6.94 is the 0.694 high-activity threshold.

Live demo: flu-surge-outlook.netlify.app

EXTENSION · WASTEWATER

Wastewater tracked flu closely but gave no early warning

2022–23

0.94

same-week correlation
Best fit: 1 week behind

2023–24

0.94

same-week correlation
Best fit: same week

2024–25

0.95

same-week correlation
Best fit: same week

2025–26

0.90

same-week correlation
Best fit: 1 week behind

National influenza A wastewater levels compared with flu-A ILI+ at lags of −4 to +4 weeks. In no season did wastewater lead; weekly changes also lined up best in the same week in all four seasons.

Conclusion: a corroborating signal, not a leading indicator, at a 1–4 week horizon.

Source: CDC NWSS; flu-A ILI+ = ILI % × % influenza A positive. Exploratory: correlations not tested for significance.

CONCLUSIONS

What we learned

1. A simple statistical model (ARIMA) matched or beat the AI model on accuracy, with better-calibrated ranges.
2. Surge probabilities are fragile: rankings flip between seasons, and only Prophet beat the naive baseline in both.
3. ILI+ gave a flu-specific, scaled target; wastewater corroborates it but does not lead.

Limitations

- Two test seasons, four threshold crossings each
- Thresholds use 2024–25 data, so that test season is not fully independent
- Revised data, not what was known in real time
- Chronos may have seen older public ILI data in pretraining
- National only; counts only people who see a doctor